

Airbnb Cape Town Business Analysis.

AUTHORS

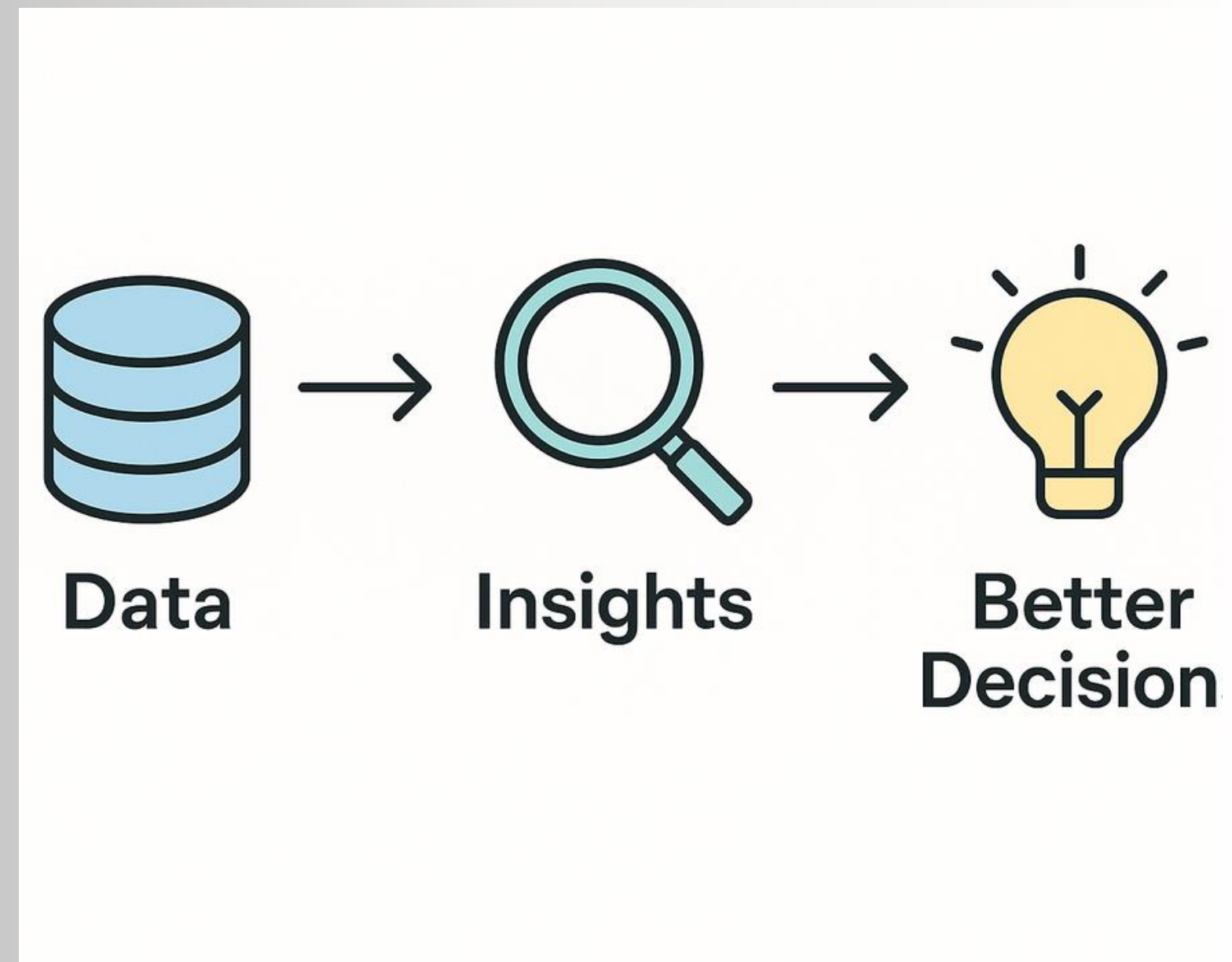
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A data-driven guide for Airbnb hosts



OVERVIEW

- Cape Town's Airbnb market is competitive and fast-growing.
- Many hosts—especially new ones—struggle to set prices and estimate revenue.
- We built a data-driven tool to help hosts:
 - Predict expected revenue.
 - Suggest improvements to increase earnings.



BUSINESS UNDERSTANDING

The Problem:

Hosts risk losing bookings or undervaluing their property without good pricing and availability strategies.

Our Solution:

A Revenue Recommender System that learns from past Airbnb data and gives clear, actionable tips.

Goal:

Help hosts maximize income while staying competitive.



Revenue

Data Sources

Data Sources

We used Airbnb's public Cape Town datasets:



Listings.csv – Property details (location, room type, amenities)



Calendar.csv – Daily prices and availability



Reviews.csv – Guest ratings and comments



Neighbourhoods.csv – Mapping of areas in Cape Town

Data Cleaning & Preparation



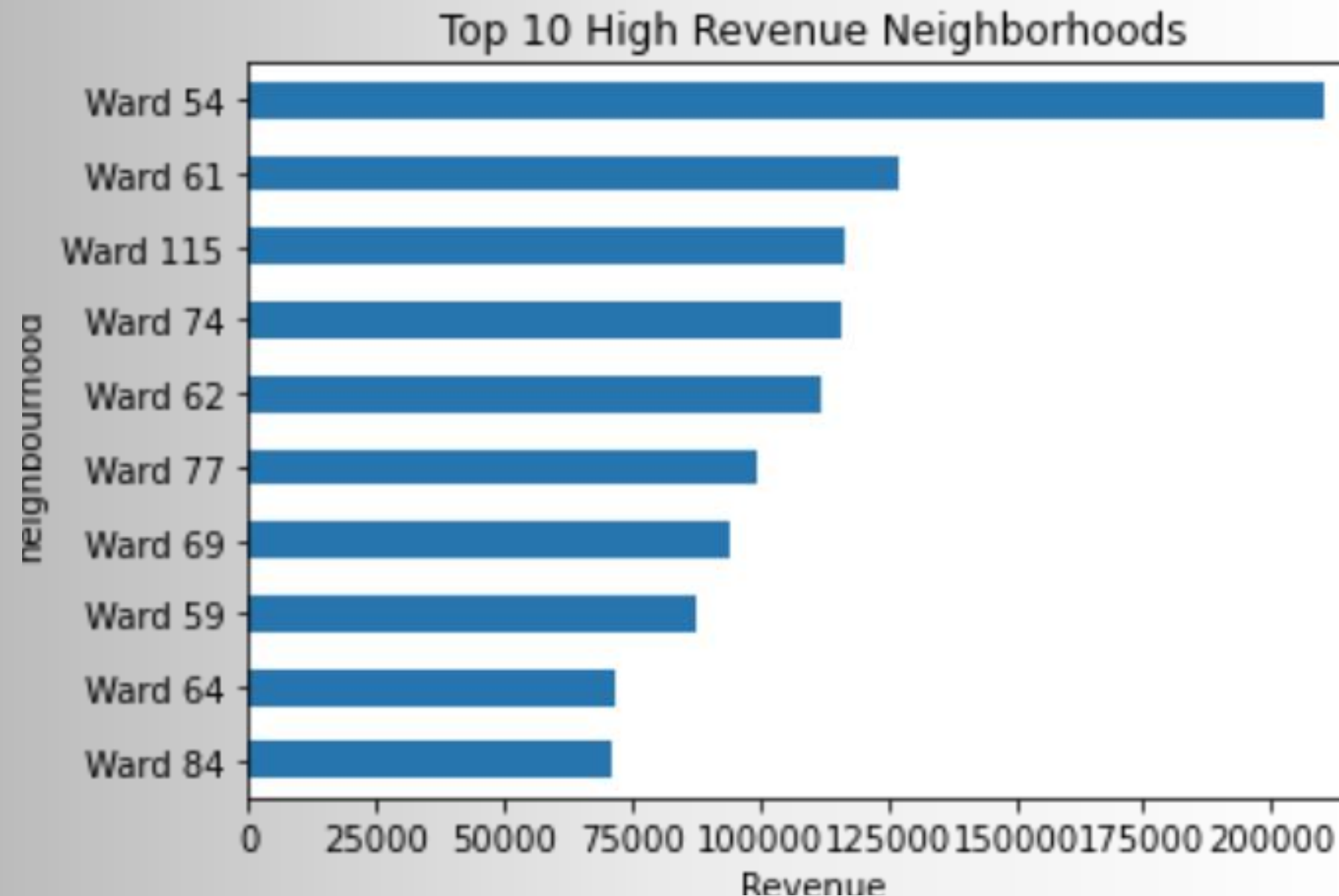
To make sure our analysis was accurate, we:

- Removed outliers (e.g., listings charging \$10,000 per night).
- Filled missing values in reviews and availability.
- Converted categories to numbers so models can understand them.
- Created new features like seasonal availability trends.

Key Data Insights

Certain neighbourhoods consistently outperform others in revenue.

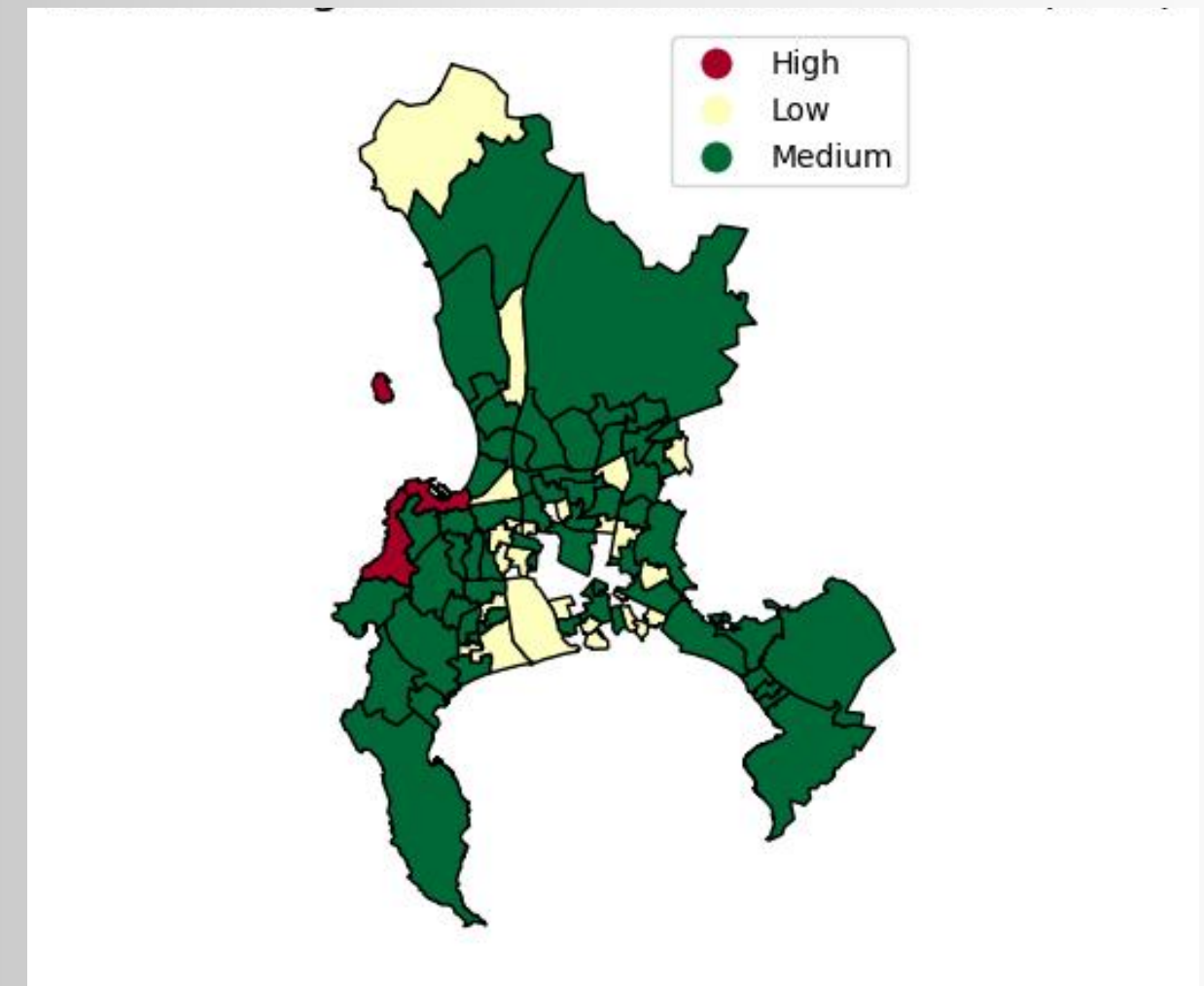
- Ward 54 is the top-earning neighbourhood, with revenue far exceeding other areas.
- Other high-revenue areas include Ward 61, Ward 115, Ward 74, and Ward 62.
- Revenue differences between wards suggest location plays a major role in Airbnb success.



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GEOGRAPHIC INSIGHTS

- Listings in tourist hotspots like the Waterfront and Camps Bay have higher prices.
 - Central locations attract business travelers, boosting occupancy.
- Distance from attractions influences revenue.



WHY MODELING HELPS

- Historical data shows patterns that humans might miss.
 - Models can predict future revenue based on listing details.
 - Think of it like a “smart calculator” that learns from thousands of past listings.



Models We Tested

We tried several “smart tools” to see which predicts revenue best:

- Simple Trend Finder (Linear Regression) – Looks for straight-line relationships between factors like price, location, and revenue.
- Pattern Finder (Random Forest) – Handles complex situations where many factors work together.
- High-Accuracy Predictor (Gradient Boosting) – Combines many small models to make very accurate predictions.
- Group Finder (Clustering) – Groups similar listings together so we can compare performance within each group.



What We Learned from Models

- Price, location, and room type are the biggest drivers of revenue
- Higher review scores boost revenue potential
- Availability patterns (especially during peak season) matter a lot



Example Scenario

Let's look at a real-life style example:

Current Situation

- A host is offering a private room.

Their property is available 50% of the time and has average guest reviews.

Predicted monthly income: \$1,500.

- If They Make These Changes:

Switch to renting the entire home.

Increase availability to 70% of the time.

Improve guest experience to get higher review scores.

- Result:

Predicted monthly income jumps to \$2,300 — that's a 53% increase.

Takeaway: Small changes in property type, availability, and reviews can make a big difference in revenue.



Recommendations for Hosts

To boost revenue, hosts should:

- Offer entire homes if possible
- Increase booking availability
- Maintain high review scores
- Optimize pricing using our model's feedback



Next Steps

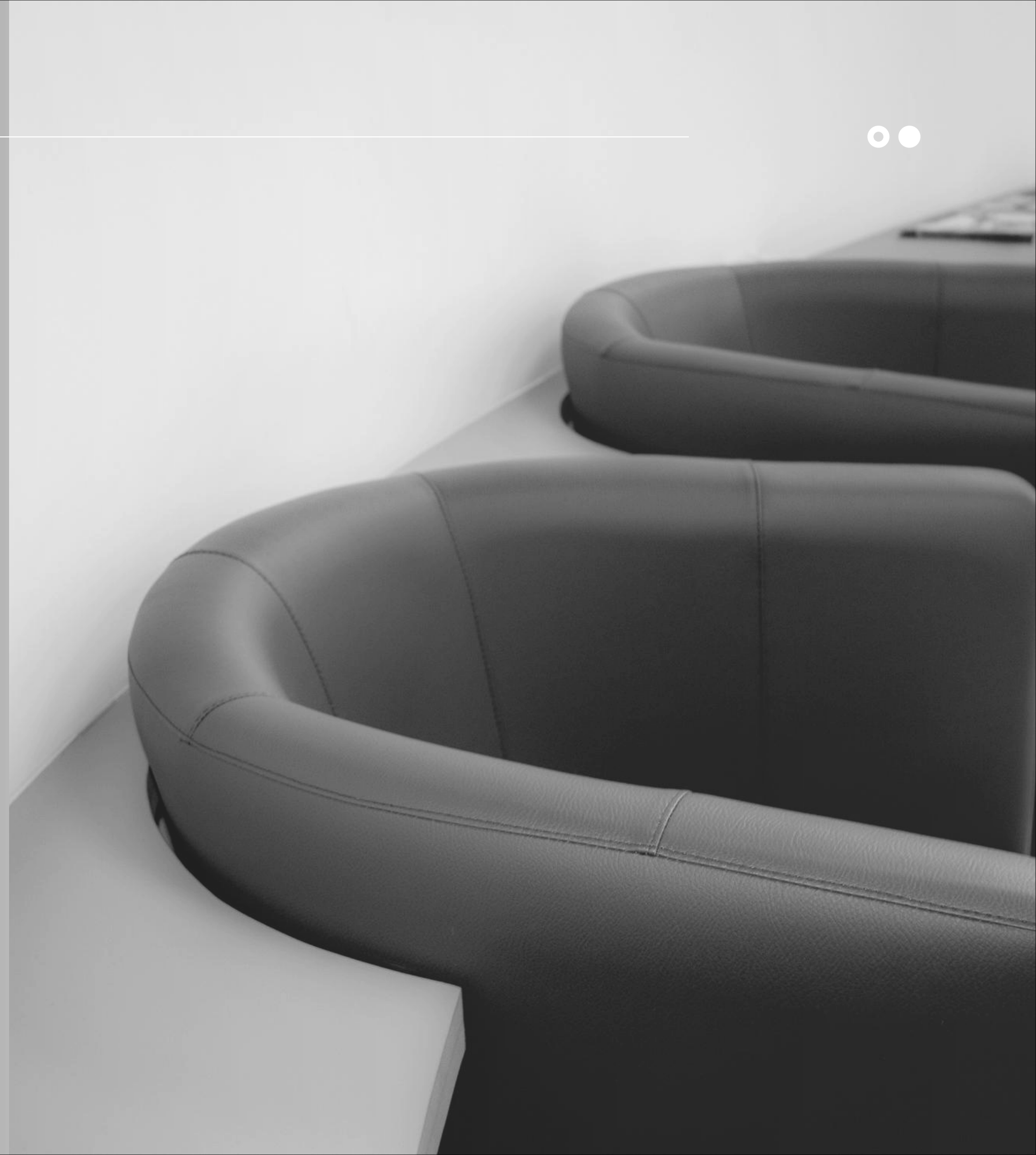
We plan to:

- ✓ Launch a prototype dashboard using Streamlit
- ✓ Allow host input + instant feedback
- ✓ Deploy model as an API for future apps
- ✓ Share with local host communities



Limitations

- Data may not reflect recent Airbnb trends
- Sparse reviews or availability can reduce accuracy
- External events (e.g., travel bans) not accounted for



Real-World Impact

- Helps hosts make smarter, data-informed decisions
- Increases revenue potential, especially for new or struggling hosts
- Can scale to other cities in the future



THANK YOU

Questions and feedback

We are ready to discuss Further